

# Deep Generative Models

## Lecture 3

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# Recap of Previous Lecture

## Jacobian Matrix

Let  $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$  be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x}), \quad \mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

## Change of Variables Theorem (CoV)

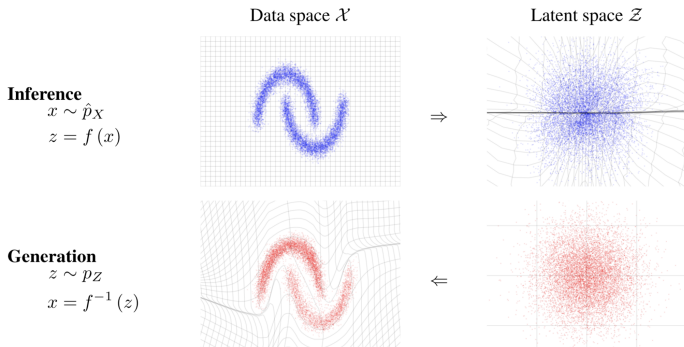
Let  $\mathbf{x} \in \mathbb{R}^m$  be a random vector with density  $p(\mathbf{x})$ , and let  $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$  be a  $C^1$ -diffeomorphism ( $\mathbf{f}$  and  $\mathbf{f}^{-1}$  are continuously differentiable mappings). If  $\mathbf{z} = \mathbf{f}(\mathbf{x})$ , then

$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left( \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left( \frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$
$$p(\mathbf{z}) = p(\mathbf{x}) |\det(\mathbf{J}_{\mathbf{f}^{-1}})| = p(\mathbf{x}) \left| \det \left( \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \right) \right| = p(\mathbf{f}^{-1}(\mathbf{z})) \left| \det \left( \frac{\partial \mathbf{f}^{-1}(\mathbf{z})}{\partial \mathbf{z}} \right) \right|$$

# Recap of Previous Lecture

## Definition

A normalizing flow is a  $C^1$ -diffeomorphism that transforms data  $\mathbf{x}$  to noise  $\mathbf{z}$ .



## Log-Likelihood

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^K \log |\det(\mathbf{J}_{\mathbf{f}_k})|$$

# Recap of Previous Lecture

## Flow Log-Likelihood

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_{\theta}(\mathbf{x})) + \log |\det(\mathbf{J}_{\mathbf{f}})|$$

One significant challenge is efficiently computing the Jacobian determinant.

## Linear Flows

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \theta = \mathbf{W}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}^{\top}$$

- ▶ LU Decomposition:

$$\mathbf{W} = \mathbf{P}\mathbf{L}\mathbf{U}.$$

- ▶ QR Decomposition:

$$\mathbf{W} = \mathbf{Q}\mathbf{R}.$$

Decomposition is performed only once during initialization. Then the decomposed matrices (**P**, **L**, **U** or **Q**, **R**) are optimized.

## Recap of Previous Lecture

Consider an autoregressive model:

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}), \quad p_{\theta}(x_j | \mathbf{x}_{1:j-1}) = \mathcal{N}(\mu_{j,\theta}(\mathbf{x}_{1:j-1}), \sigma_{j,\theta}^2(\mathbf{x}_{1:j-1})).$$

### Gaussian Autoregressive Normalizing Flow

$$\mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z}) \quad \Rightarrow \quad x_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot z_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}).$$

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) \quad \Rightarrow \quad z_j = (x_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}.$$

- ▶ This transformation is both  $C^1$ -**diffeomorphism**, mapping  $p(\mathbf{z})$  to  $p_{\theta}(\mathbf{x})$ .
- ▶ The Jacobian matrix for this transformation is triangular.

The generative function  $\mathbf{f}_{\theta}^{-1}(\mathbf{z})$  is **sequential**, while the inference function  $\mathbf{f}_{\theta}(\mathbf{x})$  is **not sequential**.

# Recap of Previous Lecture

Let us partition  $\mathbf{x}$  and  $\mathbf{z}$  into two groups:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}].$$

## Coupling Layer

$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1; \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \sigma_{\theta}(\mathbf{z}_1) + \mu_{\theta}(\mathbf{z}_1). \end{cases} \quad \begin{cases} \mathbf{z}_1 = \mathbf{x}_1; \\ \mathbf{z}_2 = (\mathbf{x}_2 - \mu_{\theta}(\mathbf{x}_1)) \odot \frac{1}{\sigma_{\theta}(\mathbf{x}_1)}. \end{cases}$$

Both density estimation and sampling require just a single pass!

## Jacobian

$$\det \left( \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & \mathbf{0}_{d \times (m-d)} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix} = \prod_{j=1}^{m-d} \frac{1}{\sigma_{j,\theta}(\mathbf{x}_1)}.$$

A coupling layer is a special instance of an gaussian autoregressive normalizing flow.

# Outline

1. Latent Variable Models (LVM) (continued)
2. Variational Evidence Lower Bound (ELBO)
3. Amortized Inference
4. ELBO Gradients, Reparametrization Trick

# Recap of Previous Lecture

## Maximum Likelihood Estimation (MLE) Problem

$$\theta^* = \arg \max_{\theta} p_{\theta}(\mathbf{X}) = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

The distribution  $p_{\theta}(\mathbf{x})$  should be highly complex (just like the true data distribution  $p_{\text{data}}(\mathbf{x})$ ).

## Latent Variable

Introduce a latent variable  $\mathbf{z}$  for each observed sample  $\mathbf{x}$ :

$$p_{\theta}(\mathbf{x}, \mathbf{z}) = p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\theta}(\mathbf{x}, \mathbf{z}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$



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$$p_{\theta}(\mathbf{x}) = \int p_{\theta}(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z}) d\mathbf{z}.$$

- ▶  $p(\mathbf{z})$  is prior distribution over the latent variable.
- ▶  $p_{\theta}(\mathbf{x}|\mathbf{z})$  is **decoder** distribution.
- ▶ Both  $p_{\theta}(\mathbf{x}|\mathbf{z})$  and  $p(\mathbf{z})$  are usually much simpler than  $p_{\theta}(\mathbf{x})$ .

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2. Variational Evidence Lower Bound (ELBO)
3. Amortized Inference
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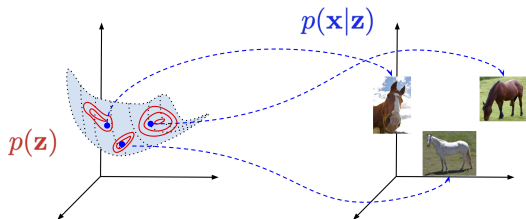
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## MLE for LVM

$$\sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i | \mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \rightarrow \max_{\theta}.$$

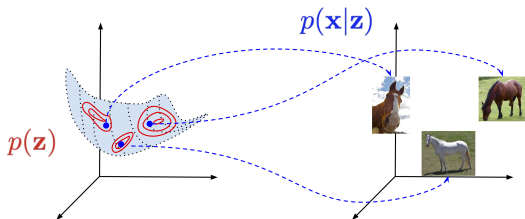
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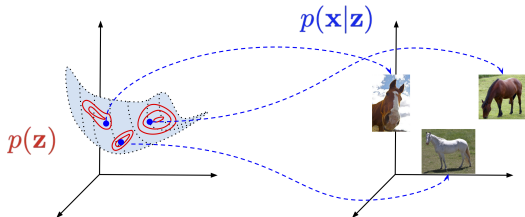
## Naive Monte Carlo Estimation

$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x} | \mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x} | \mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x} | \mathbf{z}_k),$$

where  $\mathbf{z}_k \sim p(\mathbf{z})$ .

## MLE for LVM

$$\sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i | \mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \rightarrow \max_{\theta}.$$



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$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k),$$

where  $\mathbf{z}_k \sim p(\mathbf{z})$ .

**Challenge:** As the dimensionality of  $\mathbf{z}$  increases, the number of samples needed to adequately cover the latent space grows exponentially.

# ELBO Derivation I

Bayes theorem

$$p_{\theta}(\mathbf{z}|\mathbf{x}) = \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{x})} \quad \Rightarrow \quad p_{\theta}(\mathbf{x}) = \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})}$$



# ELBO Derivation I

Bayes theorem

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Log likelihood

$$\log p_{\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z}$$

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Log likelihood

$$\log p_{\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} = \mathbb{E}_q \log \left[ \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} \right]$$

# ELBO Derivation I

## Bayes theorem

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$$\begin{aligned} \log p_{\theta}(\mathbf{x}) &= \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} = \mathbb{E}_q \log \left[ \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} \right] = \\ &= \mathbb{E}_q \log \left[ \frac{p_{\theta}(\mathbf{x}, \mathbf{z}) q(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x}) q(\mathbf{z})} \right] \end{aligned}$$

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- ▶ Here,  $q(\mathbf{z})$  is any distribution such that  $\int q(\mathbf{z}) d\mathbf{z} = 1$ .
- ▶ We assume that  $\text{supp}(q(\mathbf{z})) = \text{supp}(p_{\theta}(\mathbf{z}|\mathbf{x})) = \mathbb{R}^d$ .

# ELBO Derivation II

## Variational Decomposition

$$\log p_{\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} + \mathbb{E}_q \log \frac{q(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})}$$

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## Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\theta}(\mathbf{x})$$

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- This inequality holds for any choice of  $q(\mathbf{z})$ .

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- ▶ Instead of maximizing the likelihood, maximize the ELBO:

$$\max_{\theta} p_{\theta}(\mathbf{x}) \quad \rightarrow \quad \max_{q,\theta} \mathcal{L}_{q,\theta}(\mathbf{x})$$

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- ▶ Instead of maximizing the likelihood, maximize the ELBO:

$$\max_{\theta} p_{\theta}(\mathbf{x}) \quad \rightarrow \quad \max_{q,\theta} \mathcal{L}_{q,\theta}(\mathbf{x})$$

- ▶ Distribution  $q(\mathbf{z})$  is treated as **variational** parameter.

## Variational Evidence Lower Bound (ELBO)

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z}) \| p_{\theta}(\mathbf{z}|\mathbf{x}))$$

What is the optimal distribution  $q^*(\mathbf{z})$  given fixed  $\theta^*$ ?

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Here we got the intuition about variational distribution  $q(\mathbf{z})$  – it estimates the posterior  $p_{\theta^*}(\mathbf{z}|\mathbf{x})$ .

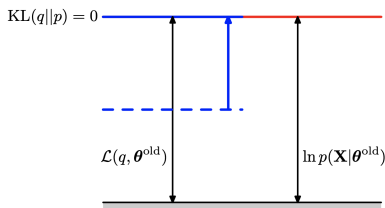
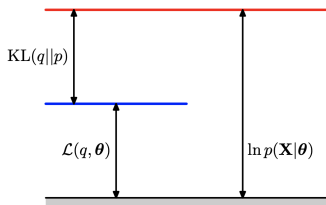
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## Variational Evidence Lower Bound (ELBO)

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$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x}))$$

# Variational Evidence Lower Bound (ELBO)

$$\begin{aligned}\mathcal{L}_{q,\theta}(\mathbf{x}) &= \int q(\mathbf{z}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} + \int q(\mathbf{z}) \log \frac{p(\mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))\end{aligned}$$

## Log-Likelihood Decomposition

$$\begin{aligned}\log p_{\theta}(\mathbf{x}) &= \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})) = \\ &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).\end{aligned}$$

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- What do we get if we drop  $D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$  term from ELBO?



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## Log-Likelihood Decomposition

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- ▶ What do we get if we drop  $D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$  term from ELBO?
- ▶ What do we have in the term  $\mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z})$  in the case of Normal distribution?

# Outline

1. Latent Variable Models (LVM) (continued)
2. Variational Evidence Lower Bound (ELBO)
3. Amortized Inference
4. ELBO Gradients, Reparametrization Trick

# Parametric Variable Posterior

## Variational Posterior

$$q(\mathbf{z}) = \arg \max_q \mathcal{L}_{q, \theta^*}(\mathbf{x}) = \arg \min_q D_{\text{KL}}(q \| p) = p_{\theta^*}(\mathbf{z} | \mathbf{x}).$$

# Parametric Variable Posterior

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- ▶  $p_{\theta^*}(\mathbf{z} | \mathbf{x})$  may be **intractable**;
- ▶  $q(\mathbf{z})$  is individual for each data point  $\mathbf{x}$ .

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## Amortized Variational Inference

We restrict the family of possible distributions  $q(\mathbf{z})$  to a parametric class  $q_\phi(\mathbf{z} | \mathbf{x})$ , **conditioned on data  $\mathbf{x}$**  and **parameterized by  $\phi$** .

# Parametric Variable Posterior

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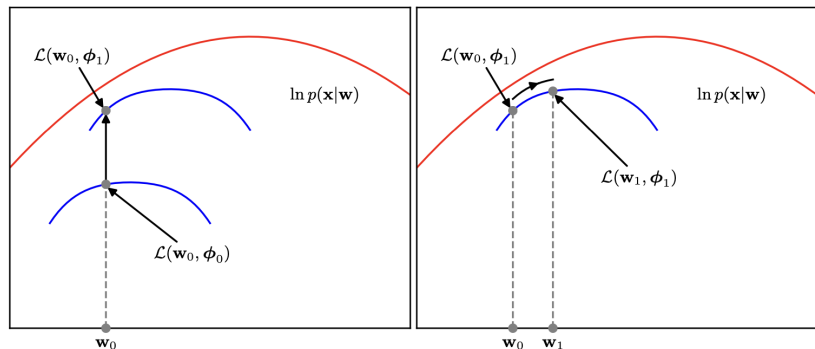
## Gradient Update

$$\begin{bmatrix} \phi_k \\ \theta_k \end{bmatrix} = \left[ \begin{bmatrix} \phi_{k-1} + \eta \cdot \nabla_\phi \mathcal{L}_{\phi, \theta}(\mathbf{x}) \\ \theta_{k-1} + \eta \cdot \nabla_\theta \mathcal{L}_{\phi, \theta}(\mathbf{x}) \end{bmatrix} \right]_{(\phi_{k-1}, \theta_{k-1})}$$

# ELBO Optimization

## Gradient Update

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# ELBO Optimization

## ELBO

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{\phi, \theta}(\mathbf{x}) + D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p_{\theta}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{\phi, \theta}(\mathbf{x}).$$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$



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- ▶  $\phi$  denotes the parameters of the variational posterior  $q_{\phi}(\mathbf{z}|\mathbf{x})$ .
- ▶  $\theta$  represents the parameters of the generative model  $p_{\theta}(\mathbf{x}|\mathbf{z})$ .

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- ▶  $\theta$  represents the parameters of the generative model  $p_{\theta}(\mathbf{x}|\mathbf{z})$ .

The remaining step is to obtain **unbiased** Monte Carlo estimates of the gradients:  $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$  and  $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$ .

# Outline

1. Latent Variable Models (LVM) (continued)
2. Variational Evidence Lower Bound (ELBO)
3. Amortized Inference
4. ELBO Gradients, Reparametrization Trick

## ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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Gradient  $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\theta} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z}$$

## ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

Gradient  $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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### Gradient $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\begin{aligned}\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \nabla_{\theta} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\ &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\ &\approx \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\phi}(\mathbf{z}|\mathbf{x}).\end{aligned}$$

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### Naive Monte Carlo Estimation

$$\log p_{\theta}(\mathbf{x}) \geq \int \log p_{\theta}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k), \quad \mathbf{z}_k \sim p(\mathbf{z}).$$



## ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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The variational posterior  $q_{\phi}(\mathbf{z}|\mathbf{x})$  typically concentrates more probability mass in a much smaller region than the prior  $p(\mathbf{z})$ .

## ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

### Gradient $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

Unlike the  $\theta$ -gradient, the density  $q_{\phi}(\mathbf{z}|\mathbf{x})$  now depends on  $\phi$ , so standard Monte Carlo estimation can't be applied:

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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## Reparametrization Trick (LOTUS Trick)

Assume  $\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})$  is generated by a random variable  $\epsilon \sim p(\epsilon)$  via a deterministic mapping  $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$ . Then,

$$\mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \mathbf{f}(\mathbf{z}) = \mathbb{E}_{\epsilon \sim p(\epsilon)} \mathbf{f}(\mathbf{g}_{\phi}(\mathbf{x}, \epsilon))$$

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**Note:** The LHS expectation is with respect to the parametric distribution  $q_{\phi}(\mathbf{z}|\mathbf{x})$ , while the RHS is for the non-parametric  $p(\epsilon)$ .

*Kingma D.P., Welling M. Auto-Encoding Variational Bayes, 2013*

# ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

## Reparametrization Trick (LOTUS Trick)

$$\nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \mathbf{f}(\mathbf{z}) d\mathbf{z} = \nabla_{\phi} \int p(\epsilon) \mathbf{f}(\mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon$$

,

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where  $\epsilon^* \sim p(\epsilon)$ .

# ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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where  $\epsilon^* \sim p(\epsilon)$ .

## Variational Assumption

$$p(\epsilon) = \mathcal{N}(0, \mathbf{I}); \quad \mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon) = \sigma_{\phi}(\mathbf{x}) \odot \epsilon + \mu_{\phi}(\mathbf{x});$$

$$q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})).$$

Here,  $\mu_{\phi}(\cdot)$  and  $\sigma_{\phi}(\cdot)$  are parameterized functions (outputs of a neural network).

Thus, we can write  $q_{\phi}(\mathbf{z}|\mathbf{x}) = \text{NN}_{e, \phi}(\mathbf{x})$ , the **encoder**.



## ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

## ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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### Reconstruction Term

$$\begin{aligned} \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} &= \int p(\epsilon) \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon \approx \\ &\approx \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \sigma_{\phi}(\mathbf{x}) \odot \epsilon^* + \mu_{\phi}(\mathbf{x})), \quad \text{where } \epsilon^* \sim \mathcal{N}(0, \mathbf{I}) \end{aligned}$$

## ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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The generative distribution  $p_{\theta}(\mathbf{x}|\mathbf{z})$  can be implemented as a neural network.

We may write  $p_{\theta}(\mathbf{x}|\mathbf{z}) = \text{NN}_{d, \theta}(\mathbf{z})$ , called the **decoder**.

## ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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### KL Term

$p(\mathbf{z})$  is the prior over latents  $\mathbf{z}$ , typically  $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$ .

$$\nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) = \nabla_{\phi} D_{\text{KL}}(\mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})) \| \mathcal{N}(0, \mathbf{I}))$$

## ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

### Reconstruction Term

$$\begin{aligned} \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} &= \int p(\epsilon) \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon \approx \\ &\approx \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \sigma_{\phi}(\mathbf{x}) \odot \epsilon^* + \mu_{\phi}(\mathbf{x})), \quad \text{where } \epsilon^* \sim \mathcal{N}(0, \mathbf{I}) \end{aligned}$$

The generative distribution  $p_{\theta}(\mathbf{x}|\mathbf{z})$  can be implemented as a neural network.

We may write  $p_{\theta}(\mathbf{x}|\mathbf{z}) = \text{NN}_{d, \theta}(\mathbf{z})$ , called the **decoder**.

### KL Term

$p(\mathbf{z})$  is the prior over latents  $\mathbf{z}$ , typically  $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$ .

$$\nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) = \nabla_{\phi} D_{\text{KL}}(\mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})) \| \mathcal{N}(0, \mathbf{I}))$$

This expression admits a closed-form analytic solution.

# Summary

- ▶ LVMs maximize the variational evidence lower bound (ELBO) to obtain maximum likelihood estimates for the parameters.
- ▶ Parametric posterior distribution  $q_{\phi}(\mathbf{z}|\mathbf{x})$  makes the method scalable.
- ▶ The reparametrization trick provides unbiased gradients with respect to the variational posterior  $q_{\phi}(\mathbf{z}|\mathbf{x})$ .